Seeing Assets Typically Unseen

Because new types of data enable new types of insights and consequently greater precision of what – and who – is likely to drive new economic value,
powered by Agentic AI and Graph Insights

A case study

Ralph Welborn, Ph.D. ralph@sunasiai.com
Vince Kasten vince@sunasiai.com
Joe Gallagher Joe@sunasiai.com

Source unique opportunities

Drive outsized returns

Deploy Capital Quickly with Precision

Context & Overview

Context: SunasiAI’s Areas of Focus

Precision insights that challenge the status quo by making visible what is typically invisible

Translation: make better decisions, faster, while synthesizing more data

1 Bets

Find potential acquisitions that others seldom see, faster than they can -

via insight into hyper-granular insights that drive economic value

2 Optimize

Unlock more portfolio value –

via hyper-granular insights into portfolio & company ‘holes’, dependencies & unlocks

3 Forecast+

Strengthen the fidelity of forecasting / monitoring / valuation –

via Bottom-up insight into the hyper-granular capabilities that drive (or destroy) economic value

4 Risk

Strengthen strategies through agentic risk profiles–

via teasing apart volatility from cyclicality indicators with clarity on ripple effects on changes in value of underlying assets / capabilities

Setting the Stage: common origination / ‘bets’ questions we answer

1

Where is disruption coming from?

1. By Capability
2. By Value chain
3. By Company

2

What companies might we consider for partnering and/or acquisition – ones that others seldom see before they do so?

1. By Capability
2. By value chain
3. By location
4. By performance

3

What capabilities are “most important” & how are they driving existing and emerging value?

1. Quadrant: changing economic value
2. Connectivity Clusters & Network Insights
3. The New 20%?

The SunasiAI difference

Let your analysts focus more time on the important targets, more effectively with greater precision

Typical Approach:
Top-Down
(yesterday's game)

1. Decide on investment thesis
2. Pick a sector / domain
3. Fill the top of the funnel through structured data typically within the sector
   (Ibankers * Consultants * 'Pitchbook-like' sources)
4. Filter via financials
5. Monitor
6. Get the same list as others

Line of Visibility stops at the company as unit of focus

Sunasi's Agentic-AI Approach:
Combining Top- Down with Bottom-Up 'Fitness' Profiles
(tomorrow's game)

1. Decide on investment thesis
2. Pick a domain / sector

3. Derive value-chain stages
4. Derive capabilities mapped to value chain stages
5. Profile 'changing economics' of value chain stages & underlying capabilities
6. Derive capability 'profile' deep-dive – the 'critical 20'
7. Derive 'fit' companies based on cap profile & criteria for focus, by value chain stage & changing economics per stage
8. Filter via financials + Moneyball-like insights
9. Get what others get BUT more importantly, get different list based on the new 20%

Line of Visibility begins at hyper-granular capabilities & technologies

The Example – deep dive to find outsized performance in a residential solar portfolio

The Example & Specific Questions Asked

The PE firm needed to find acquisition opportunities for their Renewable Energy Fund and determine how to recapitalize the business effectively.

Initial challenge	Applying the SunasiAI Capability-Driven Revolution and Bottom-Up "Fitness" approach	Results
1 Looking for acquisitions for one of their Renewable Energy Fund with a focus on Residential Solar Align the strategy across different stakeholders and figure out how to re-capitalize the business.	1. "Replaced Big 4's top-down thesis with capability-driven bottom-up analysis, revealing hidden value patterns . 2. Identify where to invest i.e. install (via a roll-up play of small firms versus manufacturing) Changing economics per value chain stage 3. What's driving those changing economics (**external factors + underlying capabilities**)	1. We parsed through more than 1,000 companies to derive a short-list of 100 companies . 2. Changed value chain focus (from 'Install' to 'Design' & 'Monitor' 3. Shifted investment thesis from "roll-up" to "innovative choke-point"
2 **Assess investment across a defined "value chain" (**of 6 stages) and validate the value in a roll up play of installers. "Locate strategic choke-points and capability clusters that drive market control" Derived 10 value chains stages (7 'core' and 3 'enabling' value chain stages).	1. Deconstructed traditional investment thesis - revealing hidden value through 1,500+ granular capabilities across value chain stages, 2. De-constructed the company into its underlying critical capabilities (9 departments, >80 functional areas and >400 capabilities) **driving each value chain stage** (via insight into clusters / dependencies / profile) 3. Isolated a set of "critical capabilities" underlying the two value chains and changing – innovative & economic profile per capability and consequently company	1. SunasIAI's analysis identified 90% of companies on their current 'watch list' of 100 companies were identified 2. Capability mapping revealed 80% of top targets were in unexpected sectors and SunasiAI completed the Deep Dive in 4 weeks. **3. "ACQUIRED 2 STRATEGIC COMPANIES FROM DIFFERENT VALUE CHAINS, CHANGING THE ECONOMICS AND COMPETITIVE POSITION WITHIN RESIDENTIAL SOLAR"**

Let's see how...

Workbench – Overview

The SunasiAI Workbench is the SaaS subscription product providing the core analytics and insights

Workbench Objective:

- Provide a 360° analysis – funds, portfolios, value chains, sectors, capabilities, companies, etc.
- Enable "moneyball-insights" from SunasiAI's underlying graph analytics, GNN, and agentic AI engines

Questions answered include

1. What comprises the extended value chain?
2. How are economics of value chain changing? What capabilities are driving those changes?
3. What is distribution of value chain by sectors and sectors by capabilities?

More than 20 pages with rich analytical insights and perspectives for the needs of different users

Sector : Capability : Company

- Filter by either to explore

Sector Composition

- Insight in terms of universe of relevant possibilities

Companies by Revenue

- Companies by Value chain, or
- Companies by location

Value Chain Focus

- Onsight into value chain profile: # sectors, # companies, # capabilities, HHI index, Moneyball Scores – to get profile of where / how economic value changing by stage

Data Browser(s)

- Capability Browser: attributes per
- Company Browser: attributes per
- Value Chain Browser: attributes per
- Fund Browser: attributes per

Workbench – deep Capabilities Profiling is at the heart of SunasiAI's value

Insight Objective:

- Isolate the **critical capabilities** to focus company & value chain focus

Questions answered

1. What **capabilities** are currently driving economic value, are cooling off and are likely to drive new economic value?
2. "Where are they": by value chain, by **"capability cluster"**? (to make sure we get bundle of capabilities needed)

Quadrant Chart

- Identifies the capabilities to focus on – given investment thesis, from perspective of what's hot, what's cooling off and what's coming next (the new 20%)

Value Chain Heatmap

- Map capabilities per value chain stage by different filters: economic value today / tomorrow; risk; sector; other
- Also, used to map companies by value chain stage with like filters

Distributions

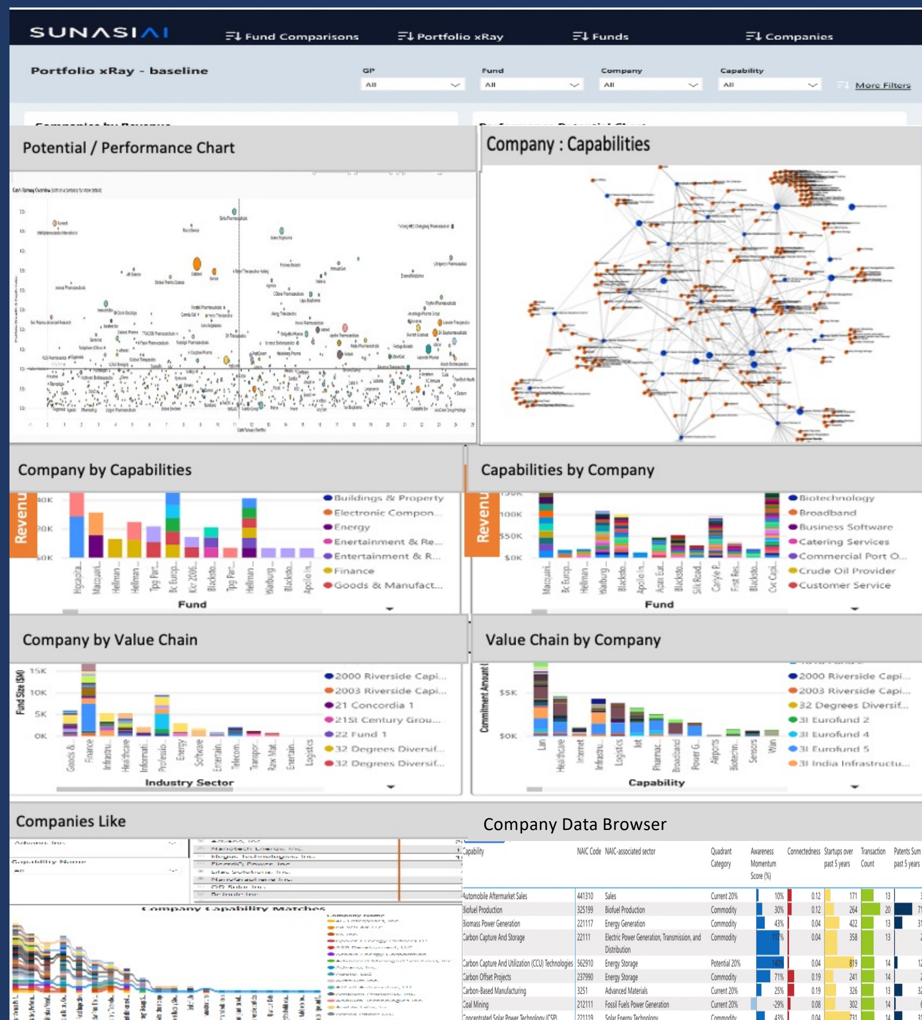
- Variety of ways to clarify where capabilities "fit" by…

Connectivity / Dependencies

Identify 'what connects with what' to help prioritize the CRITICAL 'bundle' to focus on to capture value

Also, begins to prioritize which companies "fit" given cap profile

Workbench – Companies Shortlist



Dashboard Objective:

- Shortlist candidates algorithmically to speed assessment and actions

Questions answered

1. What is the potential vs current performance profile of the companies identified (FEVI)?
2. What is the "shortlist" of companies to consider?
3. How do we sort the shortlist by different criteria – e.g., Variety of Moneyball Insights: Economic Profile, Forecasted Value, Quality of Asset, Risk Profile, Sensitivity Profile, other

2 Short-List Insights

- Performance / Performance Chart: forecasted value
- Quadrant Chart: Shifting capabilities value over time

How these matter:

1. Prioritize focus
2. See what others seldom see

Company Comparative Insights

- De-construct portcos into their underlying assets

How these matter:

1. Clarify differences
2. Isolate what actually drives value

Companies-like & Company Data Browser

- Find "like" companies based on capabilities & financial criteria

How it matters:

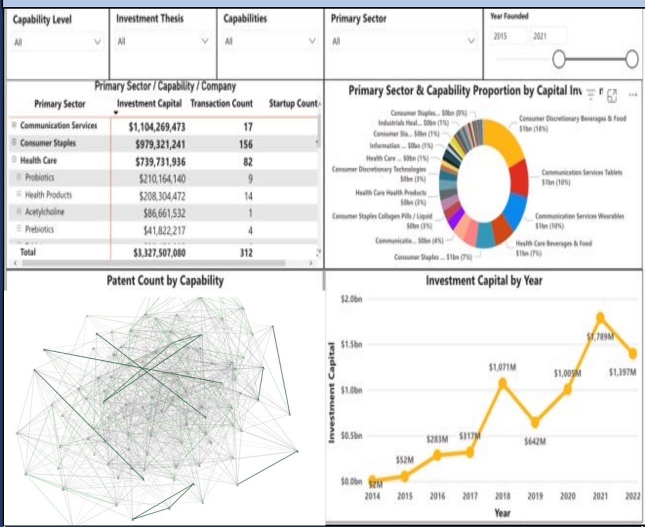
1. Speeds search for 'like" companies from Bottom-up – and financial - perspectives

Key Workbench views

Deep Dive into a few of the Dashboards

Dashboard - overview

Baseline insights

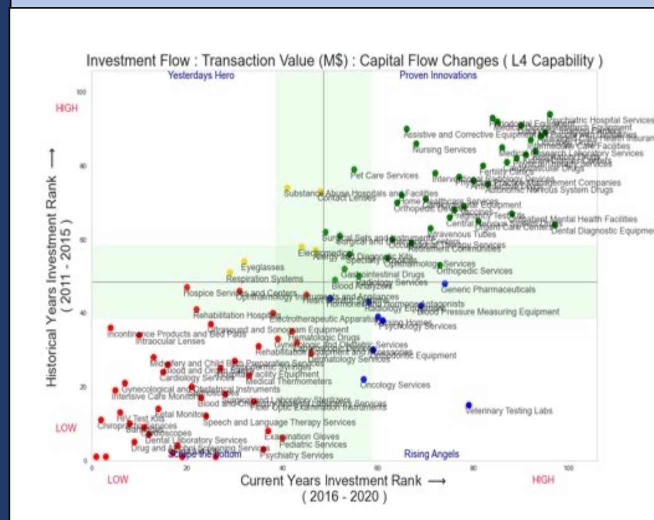


Questions answered:

1. Who are the companies that make up the universe of possible targets?
2. What is their capabilities profile?
3. How does this profile differ company by company?
4. What parts of the value chain are "hot" from investment and company focus perspectives?

Portfolio De-construction

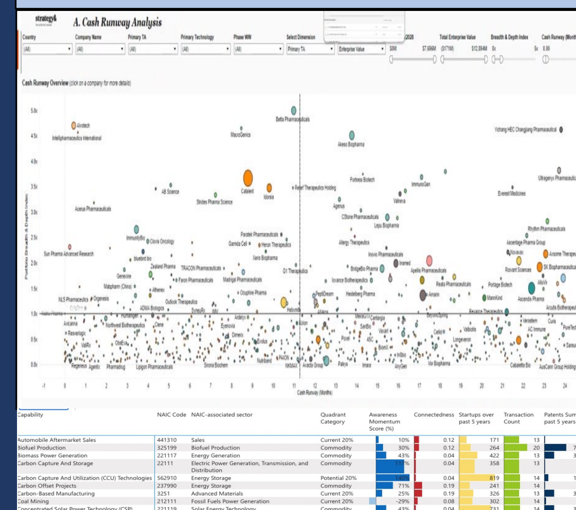
What's hot, what's not



Questions answered:

1. What is the universe of capabilities and hyper-granular technologies that make up the investment thesis?
2. Which of these are "hot", cooling off and/or promising?
3. Which companies are underpinned by these capabilities, and consequently what do I do about the insights?

Short-list



Questions answered:

1. What is the potential vs current performance profile of the companies identified?
2. What is the "shortlist" of companies to consider?
3. How do we sort the shortlist by different criteria – dozens of attributes & Moneyball-like Insights (e.g., quality of asset, economic profile, forecasted value, risk, sensitivity…)

Deep Dive into a few more

Company : Capability Mapping

Questions answered:

1. Who are the companies that make up the universe of possible targets?
2. What is their capabilities profile?
3. How does this profile differ company by company

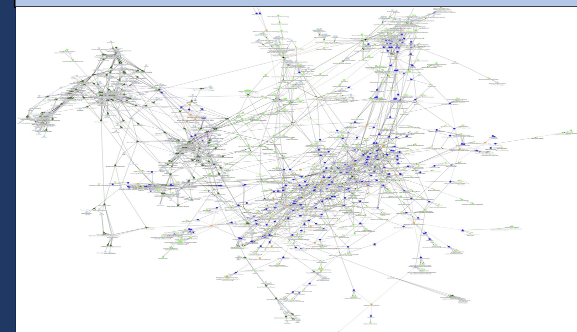
Company "Heatmap" / What lights Up – from multiple perspectives sitting on top of company 'data browser'

Capability	NAIC Code	NAIC-associated sector	Quadrant Category	Awareness Momentum Score (%)	Connectedness	Startups over past 5 years	Transaction Count	Patents Sum # past 5 years
Automobile Aftermarket Sales	441310	Sales	Current 20%	10%	0.12	171	13	32
Biofuel Production	325199	Biofuel Production	Commodity	30%	0.12	264	20	713
Biomass Power Generation	221117	Energy Generation	Commodity	43%	0.04	422	13	313
Carbon Capture And Storage	22111	Electric Power Generation, Transmission, and Distribution	Commodity	117%	0.04	358	13	29
Carbon Capture And Utilization (CCU) Technologies	562910	Energy Storage	Potential 20%	140%	0.04	819	14	125
Carbon Offset Projects	237990	Energy Storage	Commodity	71%	0.19	241	14	29
Carbon-Based Manufacturing	3251	Advanced Materials	Current 20%	25%	0.19	326	13	324
Coal Mining	212111	Fossil Fuels Power Generation	Current 20%	-29%	0.08	302	14	60
Concentrated Solar Power Technology (CSP)	221119	Solar Energy Technology	Commodity	43%	0.04	731	14	313

Questions answered:

1. What is the universe of companies (or short list of companies) per value chain stage?
2. Which companies "light up" given different criteria / filters – e.g., Quality of Asset, Risk, Forecasted Value, Economic Performance, etc.
3. How explore / filter by multiple criteria?

Underlying Graph – if want to explore the underlying data / rigor / metrics underlying insights



Cap_Name	Cap_Level	NAICS_Cod	NAICS_Cod	Market & R	Capital Mo	Cost & Pro	Economic (	Momentun	Risk Score	Sensitivity	Accelerant	FEVI Score	Grouping Score Recor
Communit	L1	624190	Other Indiv	0.83	0.85	0.85	0.84	0.81	0.83	0.88	0.85	0.83	VC: Invest
Geo-target	L1	541613	Marketing (	0.56	0.56	0.62	0.47	0.44	0.23	0.47	0.87	0.53	VC: Watch or Avoid
SMS-Based	L1	517911	Telecommu	0.53	0.32	0.42	0.42	0.64	0.36	0.16	0.81	0.46	VC: Watch or Avoid
Radio & mH	L1	515120	Television E	0.53	0.61	0.80	0.57	0.55	0.29	0.20	0.50	0.51	VC: Watch or Avoid
Health Beh	L1	541611	Administra	0.57	0.21	0.62	0.61	0.52	0.28	0.28	1.00	0.51	VC: Watch or Avoid
Maternal H	L1	611710	Educationa	0.26	0.58	0.62	0.65	0.63	0.35	0.33	0.75	0.52	VC: Watch or Avoid
Culturally /	L1	541613	Marketing (	0.24	0.67	0.35	0.66	0.43	0.31	0.43	0.93	0.5	VC: Watch or Avoid
Communit	L1	624190	Other Indiv	0.74	0.26	0.75	0.53	0.59	0.28	0.29	0.71	0.52	VC: Watch or Avoid
Multilingu	L1	541930	Translation	0.40	0.44	0.77	0.63	0.26	0.27	0.45	0.78	0.5	VC: Watch or Avoid
Interactive	L1	541511	Custom Co	0.78	0.78	0.74	0.79	0.78	0.75	0.77	0.77	0.76	VC: Invest
Digital mat	L1	518210	Data Proce	0.33	0.53	0.66	0.70	0.39	0.37	0.22	0.93	0.52	VC: Watch or Avoid
Risk stratifi	L1	541715	R&D in Phy	0.45	0.18	0.35	0.62	0.64	0.38	0.22	0.82	0.46	VC: Watch or Avoid
Geographic	L1	541370	Surveying a	0.65	0.57	0.34	0.47	0.73	0.26	0.40	0.79	0.53	VC: Watch or Avoid
Portable UI	L1	334510	Electromec	0.79	0.36	0.51	0.66	0.57	0.35	0.38	0.56	0.52	VC: Watch or Avoid
Mobile Risk	L1	541511	Custom Co	0.34	0.46	0.66	0.62	0.67	0.30	0.32	0.96	0.54	VC: Watch or Avoid
Point-of-Ca	L1	339112	Surgical an	0.84	0.86	0.84	0.83	0.90	0.84	0.84	0.81	0.85	VC: Invest
Digital Preg	L1	541512	Computer S	0.64	0.36	0.41	0.51	0.51	0.28	0.29	0.93	0.49	VC: Watch or Avoid
Anemia & I	L1	339112	Surgical an	0.69	0.70	0.50	0.28	0.48	0.20	0.26	0.73	0.48	VC: Watch or Avoid
Automated	L1	541511		0.53	0.30	0.30	0.63	0.30	0.25	0.40	0.79	0.44	VC: Watch or Avoid

Questions answered:

1. How is connected to what – as the insight foundations?
    - Capabilities : Capabilities
    - Capabilities : Companies
    - Capabilities : Value Chains
    - Companies : Companies
    - Companies : Value Chains
    - AND, all 'filtered' by multiple criteria and explorable by a set of network metrics

A bit of the method

The Residential Solar Deep Dive

Leveraging the unique capabilities of the SunasiAI workbench

We focused on the underlying capabilities of the residential solar value chain

1 **Identified 10 value chain steps (7 direct and 3 enabling) and their profile in terms of:**
- Capabilities – essential and critical enabling – for each stage of the value chain
- Companies with accretive capabilities for each value chain stage (whether in or outside of the Residential Solar market
- Sectors supporting

2 **Collected and/or derived data per company:**
- By value chain stage
- By geography
- By size
- By many other attributes

3 **Pruned list via prioritization filters** using more than 30 filters from ~5,000 to 100 including:
- Organizational
- Performance
- Market 'Fit'
- Quality of Asset
- Actionability

What we did (a subset of the value chain stages)

Extract
- Carbon Fiber-Reinforced Polymers
- Matrix Composites
- Rhodium Applications
- Waste-to-Energy Conversions and Tailings Management

Design
- Biogradable Electronics
- Mixed-Signal Designs
- Non-linear Simulations
- Radiation-Hardened Materials
- Thermal Coatings

Make
- Carbon Fiber-Reinforced Polymers
- Matrix Composites
- Rhodium Applications
- Waste-to-Energy Conversions and Tailings Management

Test
- Acoustic Microscopy / Acoustic Vibration
- External Flight Test Platform
- Quantum Error Corrections

Service
- Autonomous Maintenance
- Collaborative Robots
- Energy Harvesting
- On-Orbit Repair / Orbital Debris Tracking

Dispose
- Component Salvaging / Active Debrief Removal (ADR)
- Orbital Decay Analysis
- Space Debris Monitoring
- Space Surveillance Recovery

Examples of what we found

5 Companies
- Fiber Materials Inc.
- Ultramet,Inc.
- SGL Carbon
- …

30 Companies
- ASRC Aerospace Corporation
- Highrely Inc.

79 Companies
- API Microelectronics Liited
- Arralis Ltd
- Canopy Aerospace

8 Companies
- SARA, Inc
- Sidus Space
- Q-Ctrol
- M+P

14 Companies
- Vector Aerospace Corporation
- Turion Space

11 Companies
- RTX Corporation
- Component Recovery Solutions
- Sidus Space

Each of the 10 value chain stages were experiencing changing economic logic

- Opportunities were found to exist within these six value chain stages.
- Specific “capability bundles” were identified per stage including companies mapped to them
- The 6 stages can be “filtered down” based on a variety of criteria, including which ones are impacted the most by key structural changes or opportunities prioritized

Value Chain Stages	Extract / Source	Design	Make	Testing / QA	Services / Maintenance	End of Life / Disposal
# of Industries Providing Capabilities	14	19	27	23	17	7
Patents (centration & trend past 5 years)	882	1,231	2,494	408	293	124
Start-ups (concentration & trend past 5 years)	64	704	653	544	192	14
Capital FLow (concentration & trend past 5 years)	$2.b	$111b	$29b	$2.5b	$7b	$413m

Note: a quick breakdown of what is changing the economic logic PER STAGE also helps to pin-point where to to focus for specific opportunities, and when to do so. An example:

	Value Chain Stage & Area of Focus	*Frictions Addressed*
1	Extraction / Sourcing - New materials that form foundation for different types of products (across sectors)	Extraction economics as well as geo-political constrains of extraction and heavy reliance on a few countries for the “make” (diversification play)

Three of the Workbench core filters were used to prune the company list
from >5,000 to ~200 to focus on the "hot" capabilities across the value chain

What is the Quadrant chart?

The Quadrant chart categories hyper-granular capabilities & assets into different categories of economic contribution

Why this matters?

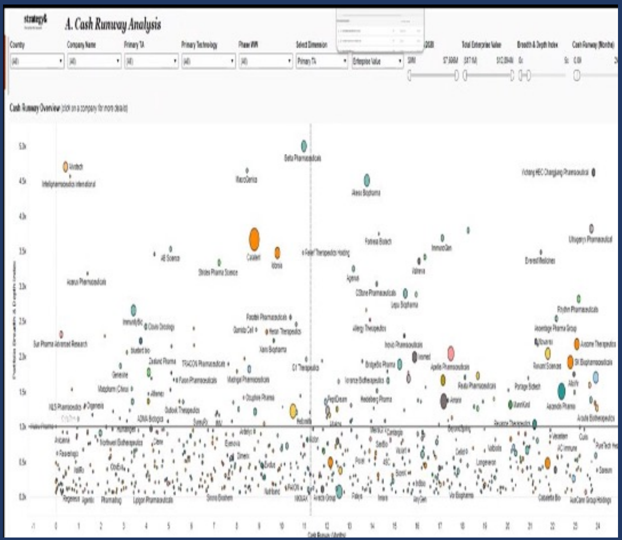
- Capabilities change in economic relevance over time.
- Consequently, what drives economic value PER company is affected.
- **Clarity of the "critical 20%" of capabilities** helps determine current & forecasted economic value.

When do we use it?

It is used to identify potential areas of differentiation and/or opportunity.

1. For any sub-sector, visualize for capabilities 'what's hot, what's not,' and then assess companies by:
   - Companies-like
   - Value chain state
   - Sector
   - Capability / offering groupings
2. Capabilities are mapped to capabilities of interest

Quadrant Insights: capabilities' economic relevance over time



What's hot, what's heating up, what's cooling down

Additional Materials

Bets – aka deal Sourcing / Origination

1

Find assets that others seldom see, quickly and at scale –

via insight into hyper-granular insights into specific hyper-granular capabilities that are creating and/or destroying economic value

Users / Role Implications

Investment Manager	Investment Manager	Investment Manager / Strategy
Use case #1: **Find distinctive deals faster, by value chain stage**	Use case #2: **Stress-test investment theses**	Use case #1: **Identify new sources of disruption – by value chain stage around which to mobilize investment**

Analyst	Analyst
Use case #1: **Assess companies more thoroughly, faster**	Use case #2: **Monitor market changes** immediately to vet if / where to respond quickly Pro-active insight into **emergent signals of risks & opportunities**

A sub-set of the Money-ball-like insights to "prune the tree": Companies

Set of filters – based on 3 different types of high-fidelity derived data: Structured, Open-Source & Synthetic

Organizational

1. **Description: biz & tech**
2. Aged Index
3. Board Members
4. Capabilities Profile
5. Capabilities Type (Type 1, Type 2)
6. Capabilities Focus (B2B, B2C, Dual Use)
7. Companies-Like
8. LP Concentration Index
9. Ownership Structure
10. Ownership Type
11. Sector / Industry
12. Year Founded
13. 10+ more

Economic Criticality

1. **Economic Criticality Score** (>10 measures)
2. Capability Concentration Index
3. Moat Score
4. Distance Score – capabilities
5. Distance Score – peers
6. Substitutability Score
7. Vulnerability Score (multiple)

Delivered as: 1) report, 2) workbook and/or 3. dashboards to interrogate

Economic Performance

1. **Revenue**
2. Capability Quadrant Score
3. Demand signals (sector)
4. Distance Scores (peer & capabilities)
5. EBITDA (if pubic)
6. Financial health (burn rates, etc.)
7. Growth
8. Headcount & HC (momentum score)
9. Investment Rounds ($ & #) Liquidity
10. Momentum Scores
11. Sector Concentration Score, comes with #4)
12. Q* score
13. Sensitivity Scores (multiple)
14. Volatility (multiple measures)
15. x/y score
16. 20+ more

Market Fit

1. **Value Chain Stages**
2. AI Disruptability Score (multiple)
3. Centrality / Criticality Scores
4. Dependencies (multiple measures)
5. Frenzy Scores
6. Market Fit Score
7. Potential score (startup, transaction, patent, grants, AI-research papers, Sectors Served *(customers) (G2, LLM)
8. Sector Uniqueness
9. Sensitivity Scores
10. 15+ more

MMM Score

1. **Top 20 Score**
2. Distance Score – Peers
3. Economic Criticality Score
4. Economic Performance Score*
5. X/Y Score (potential)
6. 3 more

Quality of Asset

1. **Moat Scores**
2. Awareness Score
3. Cap Fungibility / Reach Score
4. Companies-like
5. ESG Score (scope 3)
6. Employee / Rev Ratio
7. Risk Profile score
8. Supply Chain disruption Score
9. Sentiment : Product Sentiment : customer
10. QoA score
11. Substitutability (multiple measures)
12. Uniqueness / Distance Scores
13. 15+more

Risk Profile

1. **AI Disruptability Scores**
2. Geo-Chain Exposure
3. Substitutability Scores (multiple)
4. Supply Chain Risk Score
5. Sensitivity Scores (tariff)
6. Sensitivity Scores (interest rates)
7. Sensitivity Scores (regulatory scores)
8. Trade Profile Scores
9. >200 more

Timing / Market Readiness

1. **Actionability Score**
2. Acquisition Probability Score (ABS)
3. Correlative Timing Score
4. Exit Value Chain Stage / Alignment Score
5. Funding Rounds
6. Funding Sources
7. Last Funding Date
8. Like Acquisition Probability Score (LAPS)
9. 10+ more

Forecasted Value

1. **Q***
2. Awareness Profile (7 measures)
3. AI Disruptability Score (5 measures)
4. Capability Concentration index
5. Companies-Like Profile (5 measures)
6. Demand Signals (10 measures)
7. Dependencies / Ripple Effects (8 measures)
8. Distance Scores (multiple)
9. Frenzy Score
10. Risk Score
11. Volatility Scores (multiple measures)
12. x/y
13. 15+ more

A sub-set of the Money-ball-like Insights of filters to “prune the tree”: Capabilities

Set of filters – based on 3 different types of high-fidelity derived data: Structured, Open-Source & Synthetic

Descriptive Profile

1. **Unique ID (formal definitions)**
2. Awareness Score (2 measures)
3. Capability Quadrant Score
4. HTS Codes
5. NAIC Codes Centrality Score
6. Sentiment
7. Substitutes (8 measures)

Connectedness

1. Centrality – highly connected impacting multiple sectors
2. Betweenness – contingent (if removed, major disruptions occur)
3. Closeness – fastest reach across dependencies

Economic Profile

1. **Economic Criticality Score**
2. Capital Flow
3. Companies
4. Investment in
5. Products
6. Trade Data
7. Startups in (# & $)
8. Value Chain “placement”
9. Startup Score
10. Transaction Score
11. Q*
12. 5+ more

Economic Criticality

1. **Moat**
2. Data Intensity
3. Dependencies (see connectedness)
4. Substitutability
   Value Chain Stage(s)
5. Vulnerability – supply chain fragility (>200 measures)
6. Vulnerability – tech obsolescence (12 measures)

Quality of Asset

1. **Economic Criticality Score**
2. Data Intensity
3. Moat
4. Media Attention
5. Risk Profile Score (>200 measures)
6. Substitutability Score (>10. measures)

Risk Profile

1. **Geo_Chain (>100 measures)**
2. Regulatory Profile
3. Risk Profile + Supply Chain Fragility (>200 measures)
4. Sensitivity Scores (tariff)
5. Sensitivity Scores (interest rates)
6. Sensitivity Scores (regulatory scores)
7. Substitutability profile (>10 measures)

Sensitivity / Volatility Scores

1. **Sensitivity Scor**es (economic indicators (>250) – domain & sector specific
2. Regulatory Attention Score
3. Sensitivity Scores (tariff)
4. Sensitivity Scores (interest rates)
5. Sourcing Risk

Forecasted Value

1. **Q***
2. AI Disruptability Score (5 measures)
3. Demand Signals (10 measures)
4. Dependencies / Ripple Effects (8 measures)
5. Distance Scores (multiple)
6. Frenzy Score
7. Regulatory Profile (8 measures)
8. Startup Score
9. Substitutability (8 measures)
10. Value Chain Measures (changing economic logic (10+ measures)
11. 10+ more